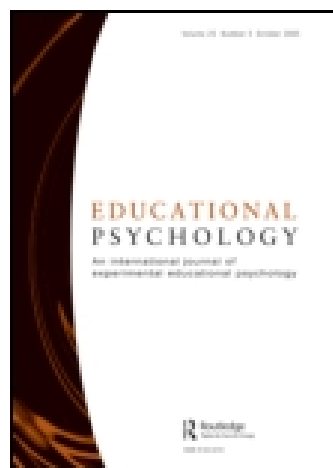


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Relations among Epistemological Beliefs, Academic Achievement, and Task Performance in Secondary School Students

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Relations among Epistemological Beliefs, Academic Achievement, and Task Performance in Secondary School Students

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Students with differing profiles of epistemological beliefs—their beliefs about personal epistemology, intelligence, and learning—vary in thinking, reasoning, motivation, and use of strategies while working on academic tasks, each of which affect learning. This study examined students' epistemological beliefs according to gender, school orientation, overall academic achievement, and performance on two differently structured academic tasks. Epistemological beliefs in fixed and quick ability to learn, simple knowledge, and certain knowledge differed significantly as a function of gender, school orientation, and levels of academic achievement. These beliefs, particularly the belief in simple knowledge, significantly predicted overall performance and reflective judgment scores on the ill-structured task but not on the well-structured task. Implications concerning the relations among epistemological beliefs, reflective judgment, gender, school orientation, task structure, and achievement are discussed.

Among a number of influences on student learning and performance are students' personal epistemologies—their beliefs about the nature of knowledge. Psychologically and educationally, personal epistemology beliefs are “how individuals come to know, the theories and beliefs they hold about knowing, and the manner in which such epistemological premises influence the cognitive processes of thinking and reasoning” (Hofer & Pintrich, 1997, p. 88). For example, some learners believe that knowledge is simple (unrelated facts) and certain (absolute), and that learning is a genetically inherited disposition that occurs quickly (Schommer, 1990, 1993), and reason simplistically, with inadequate justification, from an individual frame of reference (King & Kitchener, 1994). In the last decade, the construct of personal epistemology

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has lacked clarity in its name, definition, and content. For example, some (e.g., Schommer, 1990, 1993; Schommer-Aikins, Duell, & Barker, 2002) have used the term “epistemological beliefs” and included beliefs about intelligence and learning in with a more philosophical notion of epistemology centered on beliefs about knowledge and knowing. Following such precedent, the term “epistemological beliefs” is used in this paper to refer to a construct which includes beliefs about personal epistemology, intelligence, and learning.

Among the correlates of epistemological beliefs are students’ academic achievement (Schommer, 1993; Wood & Kardash, 2002), motivation (Hofer, 1999), cognitive development (Boyes & Chandler, 1992), skills in argumentation (Kitchener & King, 1981), conceptual change (Mason & Boscolo, 2004), comprehension of written (Kardash & Scholes, 1996) and mathematical text (Schommer, Crouse, & Rhodes, 1992), and self-efficacy (Bandura, 1993). Various profiles of personal epistemology beliefs have also been shown to handicap students’ comprehension and cognition while performing learning tasks in school (Qian & Alvermann, 1995; Schraw, Dunkle, & Benedixen, 1995). This study is an attempt to explore relations between students’ epistemological beliefs and gender, school orientation, academic achievement, and performance on two differently structured learning tasks.

Epistemological Beliefs

Conceptualisations of epistemological beliefs reflect different philosophical views held about the origins of scientific knowledge and learning. Early theories of personal epistemology characterised the construct as relatively unidimensional—having a rather uniform developmental trajectory. These include Perry’s (1968) scheme of intellectual and ethical viewpoints, ways of knowing (Belenky, Clinchy, Goldberger, & Tarule, 1986), epistemological reflection (Baxter Magolda, 2002), epistemic doubts (Boyes & Chandler, 1992), and persuasive argumentation (Kuhn, 1991). Several recent theories have stressed the multidimensionality and variable developmental trajectories of personal epistemology as beliefs about schooled knowledge (Buehl, Alexander, & Murphy, 2002), epistemic beliefs (Schraw et al., 1995), and reflective judgments (King & Kitchener, 2002; Kitchener & King, 1981).

A general theme in current epistemological belief theory postulates that students develop from a position where knowledge is “perceived as right or wrong to a position of relativism and then to a position in which individuals are active constructors of meaning, able to make judgments and commitments in a relativistic context” (Hofer & Pintrich, 1997, p. 121). Theorists (e.g., Hofer, 2001) elaborated this to characterise epistemological belief development as a continuum. Learners initially hold a dualistic view of knowledge as certain and objective. This evolves to a position where knowledge is considered relative, and multiple but inconsistent positions can be held simultaneously: this is called uncertain subjectivism. Ultimately, learners can acquire the capability to coordinate multiple and contrasting views relative to context while grounding evaluation of validity in relation to context.

The developmental sequence in many of the models of personal epistemology takes place in different stages, phases, or perspectives, generally climaxing with a cognitively mature individual who “thinks epistemologically, relativistically, and dialectically—thinking that goes beyond formal logic beyond formal operations” (Schommer, 1994, p. 308). For example, Perry’s (1968) intellectual and ethical viewpoints scheme has nine stages; Baxter Magolda’s (2002) epistemological reflection model aligns four ways of knowing with Perry’s (1968) scheme; King and Kitchener’s (1994) reflective judgment model uses interviews in which ill-structured problems are posed and can have seven stages of response; and Kuhn’s (1991) model examining persuasive argumentation on ill-structured problems posits three categories of epistemological views (absolutist, multiplist, and evaluativist).

The most widely used quantitative approach to assessing beliefs about the nature of knowledge and of learning in educational research has been Schommer’s (1990) 63-item Epistemological Questionnaire (SEQ) and her Modified Epistemological Questionnaire (SMEQ) for high school students (Schommer, 1993; Schommer & Dunnell, 1997). In it, items are categorised into 12 subsets based on the view that epistemological beliefs are relatively independent (multidimensional) rather than part of a coherent progressive structure. For example, epistemological beliefs can vary within individuals along various dimensions or factors. Schommer (1990) initially theorised five dimensions of epistemological beliefs, but results from administrations of her questionnaire have identified only four: speed of acquiring knowledge (learning quickly or not at all), simplicity of knowledge (isolated facts to memorise or highly interrelated concepts to understand), certainty of knowledge (absolute rather than contextual), and sources of knowledge and ability (innate rather than acquired). Viewing knowledge as simple, certain, or innate reflects a person’s beliefs about the nature of knowledge, while views on the speed of knowledge acquisition and the source of ability signal a person’s beliefs in the nature of learning. Her fifth theorised epistemological belief component (belief in omniscient authority or that knowledge is handed down by authority rather than attained through reason) has not been consistently identified in subsequent research (e.g., Qian & Alvermann, 1995) using the SEQ or SMEQ.

Research using the SEQ and the SMEQ has revealed links between several of Schommer’s proposed dimensions and academic achievement. For example, in several studies involving high school students, belief in quick learning has predicted grade point average (Schommer, 1993; Schommer, Calvert, Gariglietti, & Bajaj, 1997), while beliefs that knowledge is simple and certain and that learning occurs quickly have predicted conceptual change (Qian & Alvermann, 1995). Schommer (1990; see also Schommer et al., 1992) has found relations between certain epistemological beliefs and accuracy of comprehension calibration, gender (1993), critical thinking and integration of information (Schommer & Dunnell, 1997), coping with complex tasks and using strategies for studying (Schommer et al., 1992), achievement (1990, 1993; Schommer et al., 1992), students’ age and year in school (Schommer et al., 1997), and parental educa-

tion and occupational prestige (1990). Epistemological beliefs are also salient in a variety of instructional contexts (Hofer, 1994, 1999, 2001; Mason & Boscolo, 2001) and domains (Buehl et al., 2002; Hofer, 2000; Schommer-Aikins et al., 2002). An interesting though as yet untapped area of investigation involves how religious views might influence epistemological beliefs and subsequent academic achievement.

In the few studies investigating gender differences in epistemological beliefs, female students had significantly fewer debilitating beliefs in quick learning (Enman & Lupart, 2000; Schommer, 1993; Schommer et al., 1997; Wood & Kardash, 2002), in fixed ability (Enman & Lupart, 2000; Schommer, 1993; Schommer et al., 1997), in certain knowledge, and in authority and expertise as the source of knowledge (Hofer, 2000). However, Wood and Kardash (2002) found that college males had more sophisticated beliefs about the structure, construction, and modification of knowledge, with stronger beliefs "that knowledge is often complex, interrelated, and ambiguous with the implication that sometimes there is no one right answer" (p. 250). Other studies (e.g., Conley, Pintrich, Vekiri, & Harrison, 2004) have not found gender differences in epistemological beliefs. More research is needed into gender differences on various dimensions of epistemological beliefs.

Epistemological Beliefs and Task Performance

Research indicates that individual differences among students mediate how features of tasks relate to learning and performance (Blumenfeld et al., 1991; Lodewyk & Winne, 2005). For example, Balzer, Doherty, and O'Connor (1989), along with Schommer et al. (1992), reported that students who believed a task or concept was "just a bunch of isolated facts" were less inclined to select tactics for learning relationships within and among concepts. Researchers have asserted that some profiles of epistemological beliefs can bias learners' perceptions of important conditions in academic domains (Buehl et al., 2002; Hofer, 1999) and tasks (Hofer, 2001; Winne & Hadwin, 1998), and may interact with related behaviours like helplessness (Pintrich, Marx, & Boyle, 1993) or constructs like belief in fixed (innate) ability (Dweck & Leggett, 1988) when learners engage in difficult tasks. Schoenfeld (1985) reported that the more students believe learning is quick, the less time and effort they will invest in problem-solving.

Over time, the structure of learning tasks may shape epistemological beliefs that are difficult to change. For instance, in his theory of epistemic motivation, Kruglanski (1990) asserts that learning in the context of ambiguous tasks (Doyle, 1983) is shaped by both cognitive and motivational factors such that students either seek closure (cognitively freeze) or avoid closure (cognitively unfreeze). In reviews, Hofer (2001) and Hofer and Pintrich (1997) reported that students who avoid closure more actively engage in hypothesis testing, thereby creating opportunities to progress along the developmental continuum of epistemological beliefs about knowledge. In this context, epistemological beliefs may play a key role as standards or

goals for learners to calibrate comprehension (Pintrich et al., 1993) and evaluate feedback as they self-regulate learning (Butler & Winne, 1995).

In various lines of research, epistemological beliefs have been assessed in the context of students pursuing ill-structured or well-structured tasks (King & Kitchener, 1994; Schraw et al., 1995) or domains (Buehl et al., 2002; Hofer, 1999). Frederiksen (1984) and others (e.g., Spiro, Coulson, Feltovich & Anderson, 1988) theorise that learning tasks generally display both well-structured and ill-structured features. A relatively well-structured task (WST) is usually more clearly formulated and presented, and requires straightforward operations, providing the necessary information, algorithms, and precise criteria to indicate how the completed learning task will be assessed to aid the learner in searching for a suitable answer (Frederiksen, 1984; Schraw et al., 1995). A relatively ill-structured task (IST) makes less obvious the operations to be used, is more ambiguous (does not have a clearly identifiable answer), and does not provide the necessary information (clear instructions), algorithms (cues and resources), or evaluative criteria for students to determine if they are solving the problem correctly (Lodewyk & Winne, 2005; Spiro et al., 1988).

A task's positioning on this continuum of well-structured to ill-structured can have important effects. King and Kitchener (1994) reported on five studies ($n = 172$) of the reflective judgment abilities of high school students completing ill-structured problems. Students consistently demonstrated pre-reflective thinking (Stages 1–3 of 7; mean = 3.2) that indicated beliefs in certain knowledge and decision-making about tasks, while relying mainly on personal belief and feeling rather than on sound argumentative reasoning and justification, which would have indicated more quasi-reflective (Stages 4 and 5) and reflective thinking (Stages 6 and 7). However, Blumenfeld et al. (1991) noted that tasks which are not developmentally appropriate, have stringent time limits, have only one correct answer or process, are performed in an environment that focuses on performance at the expense of mastery, and are inauthentic are also probably correlated with less productive profiles of epistemological beliefs. In essence, overly well-structured tasks can theoretically curtail self-regulation (Doyle, 1983), depress learning and interfere with transfer (Spiro et al., 1988), limit self-determination (Pintrich et al., 1993), and lower performance (Schommer, 1990, 1993), particularly in students who are learning new content and have low domain knowledge (Kanfer & Ackerman, 1989).

For these reasons, some researchers (e.g., Blumenfeld et al., 1991; Doyle, 1983; Frederiksen, 1984; Spiro et al., 1988) are concerned about the high proportion of overly well-structured tasks (like repetitive recall worksheets emphasising low-level facts) in school. These researchers observe that people are much more likely to face ill-structured tasks outside school. However, overcompensating by assigning too many ill-structured tasks in school might also prove counterproductive. Students' anxiety levels climb in the face of such ambiguous and cognitively challenging tasks (Butler & Winne, 1995; Doyle, 1983). A balance of tasks seems necessary to facilitate more cognitive activity and conceptual change in students.

Research Questions and Hypotheses

Despite some research signalling that students with differing profiles of epistemological beliefs vary in thinking, reasoning, use of strategies, and learning while working on tasks (e.g., Kitchener & King, 1981; Schommer & Dunnell, 1992), relations between epistemological beliefs and school orientation (religiosity), levels of academic achievement, gender, reflective judgments, and more fine-grained instructional features like the structure of specific academic tasks have yet to be discovered.

The aim of this study is to illuminate any such relations via the following seven specific inquiries using data collected in a regular school setting.

1. An exploratory factor analysis of SMEQ items was undertaken, and the dimensions found are reported.
2. Since there is no known research addressing differences in epistemological beliefs according to school orientation (independent-religious and public schools), such results are described.
3. As tested in previous research (e.g., Kardash & Scholes, 1996; Schommer, 1990, 1993; Schommer et al., 1992), the hypothesis that epistemological belief scores should predict students' overall academic achievement is assessed.
4. A study of gifted high school students (Schommer & Dunnell, 1997) has shown links between fixed ability beliefs and academic performance; however, evidence relating epistemological beliefs to students' levels of overall academic achievement (high, moderate, or low) is still scant. In the fourth area of inquiry, a strong hypothesis is waived in favour of a descriptive exploration of potential differences in epistemological beliefs according to levels of overall academic achievement.
5. As articulated in the theoretical review, a learner's profile of epistemological beliefs may differ as a function of the characteristics of different learning tasks. This is examined. It is hypothesised that epistemological beliefs will have a stronger predictive relationship with students' performance on an ill-structured task than on a well-structured task. This hypothesis is justified by research (e.g., Schommer, 1990; Schommer et al., 1992; Schraw et al., 1995) discovering relations between epistemological beliefs in students and their persistence, engagement, and coping skills in ill-defined domains. Since ill-structured tasks are probably perceived by students to be more cognitively demanding than tasks that are highly structured, naïve (non-availing) epistemological beliefs, particularly perceptions that ability is fixed and that knowledge is simple, are expected to have a stronger mediating and predictive relationship with performance on a loosely structured task than a more highly structured task.
6. Since an ill-structured task demands that students judge reflectively, achieve academically (King & Kitchener, 1994, 2002), and argue persuasively (Kuhn, 1991), this study endeavours to illuminate further the relations between reflective judgment and achievement on an ill-structured task by testing the assertion that epistemological beliefs and levels of overall academic achievement will predict students' reflective judgment scores on an ill-structured task. For example,

students believing that knowledge is certain, an accumulation of concrete facts, and that learning is not an acquired process are predicted to perform worse on an ill-structured task than those reporting beliefs that knowledge is more ambiguous, worth questioning, and complex, and that learning is a gradual process requiring time and effort. Lower academic achievers are expected to have less ability to self-regulate their learning and performance and thus less aptitude for engaging in quasi-reflective judgment (King & Kitchener, 2002).

7. Gender differences in epistemological beliefs are tested relative to task performance, overall academic achievement, and reflective judgment, to ascertain whether girls believe less in quick learning and fixed ability than boys (Enman & Lupart, 2000; Schommer, 1993; Schommer et al., 1997) and whether girls have stronger reflective ability, as found by Wood (1993).

Method

Procedure and Participants

Combined sample. To derive a statistically robust factor structure for subsequent analyses, a sample of 447 volunteer students (79%) from 10th grade science classes in six secondary schools (five public and one independent) in a large city in western British Columbia completed a slightly modified version of Schommer's (1993) Modified Epistemological Questionnaire for high school students (SMEQ). They also completed a demographic questionnaire (DQ), in which they reported estimates of their academic average in the 9th and 10th grades. The first author and the classroom teacher administered each measure during regular class time. The sample included 219 males (49%) and 228 females (51%) from lower to upper middle-class socio-economic levels; the mean age was 15.3 years; 85% were Caucasian, 9% Asian-Canadian, and 6% other.

Public school sub-sample. Of the 447 students in the combined sample, 358 were enrolled in public schools. These included 174 males (48.6%) and 184 females (51.4%), from lower to upper middle-class socio-economic levels; the mean age was 15.2 years; 86% of these students were Caucasian, 7.5% Asian-Canadian, and 6.4% other.

Independent school sub-sample. During regular school hours, 94 students (included in the combined sample) from all four 10th grade science classes at an independent (religious) secondary school completed the same questionnaires (SMEQ, DQ), and also volunteered to complete two differently structured tasks (and another questionnaire not used in this study) as reported in Lodewyk and Winne (2005). A pre-requisite to enrolment in this school was adherence to the Protestant-Christian religious faith. All of the students in each of the classes agreed to participate in the study, although two students were excluded because of a lack of class attendance and three were dismissed

for not submitting the completed task. Of the remaining 89 students 45 were male (50.6%) and 44 were female (49.4%); they were mainly middle class; the mean age was 15.3 years; 82% were Caucasian, 14.6% Asian-Canadian, and 3.3% other.

The administration of tasks was counterbalanced to avoid order and time-of-day effects. Half of the classes first completed an ill-structured task; the remaining classes first completed a well-structured task (administered by the classroom teacher). Following completion of this task, students completed the other task. Students were given eight consecutive regularly scheduled 80-minute class periods (in classes meeting on alternate days) to complete both tasks (four for each task). Care was taken to ensure that equal instructional cues were given to each class. Students that had not completed typing up the task response by the end of the fourth and eighth periods (approximately 50%) were permitted to take their work files home to finish typing them and submit them at the beginning of the next class. For more information about these tasks, please see Lodewyk and Winne (2005).

Measures

Overall academic achievement. All students reported their estimates of overall academic achievement (EOAA) on the DQ, which was designed by the first author. It consisted of 14 brief fill-in-the-blank questions asking students to report their name, age, gender, residence, any disability, study habits, and their EOAA in the 9th and 10th grades (the DQ was administered in late March). Students were categorised into high, moderate, and low groups according to their EOAA to ensure relatively equal sample size and variance for the high and low groups. Students in the high EOAA group ($n = 124$) reported grades above 80% while those in the low EOAA group ($n = 124$) reported grades below 70%. The moderate EOAA group ($n = 199$) comprised the remaining students.

To test the concurrent validity of the EOAA, independent school students' actual overall academic achievement (OAA) in Grade 9 and Grade 10 was provided by the school. Then an overall weighted score was computed to reflect achievement over time, taking 40% of their previous year's overall academic average and 60% of the mean of their two most recent academic terms. The means for students' reported EOAA and calculated OAA were 75.66 (8.90) and 74.43 (8.81); the Pearson correlation coefficient between them was .91. This coefficient provides ample leverage for using EOAA as a valid measure of academic achievement.

Epistemological beliefs. Students' epistemological beliefs were assessed using Schommer's (1993; Schommer et al., 1992, 1997; Schommer & Dunnell, 1997; see also Qian & Alvermann, 1995) SMEQ for high school students. This questionnaire is a self-report instrument also previously used to assess college students' responses to 63 reflective statements about knowledge and learning (e.g., Enman & Lupart, 2000; Jeung, Johnston & Anderson, 1993; Kardash & Howell, 2000; Schommer, 1990; Wood & Kardash, 2002). Schommer's MEQ and SMEQ (or variations thereof) are the most commonly used self-report measures (Hofer, 2005).

Students rate themselves on a five-point Likert scale ranging from “strongly disagree” (1) to “strongly agree” (5). Some of the items were: “To be successful in school, it is best not to ask too many questions,” “You can believe almost everything you read,” “Once a person is born, their ability to learn will not change,” and “Successful students understand things quickly.” Schommer (1990, 1993; Schommer et al., 1992, 1997; Schommer & Dunnell, 1997) clustered items into 12 subsets reflecting degrees of student naïveté about seeking single answers, avoiding integration and ambiguity, criticising authority, learning how to learn, and succeeding from hard work and concentrated effort.

Previous studies using the SMEQ (e.g., Schommer, 1993; Schommer et al., 1997) report moderate test-retest reliability (.74), internal consistency coefficients for subsets in each factor of .63–.85, and reasonable predictive validity. For instance, several studies (Schommer, 1993; Schommer et al., 1992, 1997) demonstrate strong stability in the simple knowledge factor and consistent subset loading onto hypothesised factors from several subsets. Despite its popularity, however, the factor structure of the SMEQ frequently deviates from the pattern of loadings Schommer (1990, 1993) originally hypothesised. This may be because of contextual differences and covariance among items (Hofer, 2001; Hofer & Pintrich, 1997). To account for these concerns, an exploratory factor analysis of items was administered to delineate the major dimensions of the factor structure within the combined sample.

Results of a pilot study with 24 10th-grade science students from a second independent school were used to modify the SMEQ for the age and comprehension level of students in this study. For example, some statements were changed to enable 10th-grade students to comprehend and answer the items more easily. Words like “professor,” “ultimately,” “intent,” and “original” were replaced with “teacher,” “eventually,” “purpose,” and “creative.” The wording of some other questions was changed to reduce some anticipated cultural differences for the Canadian sample used in this study. For example, “Nothing is certain, but death and taxes” was replaced by, “We can’t be sure of anything, except that we will die someday.” One question (“A clean desk is a sign of an empty mind”) was omitted because the majority of students reported difficulty comprehending it even after it was revised. An additional 10 items that were theorised by Schommer (1989) to load on the omniscient authority factor were also deleted from subsequent analyses, as in previous research (Qian & Alvermann, 1995), because the factor has not been consistently identified in previous research (Schommer, 1990, 1993; Schommer et al., 1992).

Learning tasks. Both learning tasks related to cancer and were designed by us with close consultation with the classroom teacher. Existing theoretical and empirical research guidelines (e.g., Doyle, 1983) were implemented in structuring the tasks. Each task was created to resemble task structures assigned regularly in actual classroom settings (Blumenfeld et al., 1991; Doyle, 1983; Frederiksen, 1984; Spiro et al., 1988), and cues inherent in each task’s structure were varied for students to self-regulate their learning and performance. Each task was also purposely structured

“moderately well” or “moderately ill” according to its degree of complexity, structural presentation, available resources, instructional cues, expectations, and criteria for grading.

The well-structured task was entitled “Personal Lifestyle Plan.” It was designed to prompt students to gather information and gain knowledge about a chosen form of cancer. Students were required to develop a typed essay with information about the cancer’s causes, risk factors, symptoms, treatments, and potential preventative behaviours to reduce risk. To stimulate self-reflection on one’s risk for cancer, the well-structured task included an advance organiser. For example, prior to being introduced to the task, students were shown a 20-minute videotaped program advocating a lifestyle of cancer prevention and were given a worksheet to complete. To structure the task further, specific objectives (sub-goals) for completing the task (format, style, content) were presented in the task description, cancer-specific resources were given to students in pre-packaged files, and grading criteria was clearly specified.

In contrast, the ill-structured task was entitled “Task-Force Issue” and was designed to stimulate more critical thinking through argumentation about whether government funding for cancer should focus more on preventative efforts or treatment programs. In essay form, students had to justify how the government should respond to cancer based on perceived priorities for battling it. This task was structured to require students to address a more ambiguous and complex cognitive problem, and engage in more reflective judgment (King & Kitchener, 2002) and epistemological reflection (Baxter Magolda, 2002). It included fewer instructional cues, required students to find their own resources in the classroom, and had less detail and evaluative criteria in the outline for judging progress. For more information about these tasks, please see Lodewyk and Winne (2005).

Learning task performance. Performance on the well-structured and ill-structured tasks was assessed using a composite mark based on the grade assigned by the first author using a scoring rubric (weighted .90) combined with scores on a 10-item multiple-choice quiz (weighted .10) that students completed after each task. The well-structured task was graded according to how well students fulfilled requirements outlined in the rubric provided in their original task descriptions. To grade the ill-structured task, the essay scoring criteria from the British Columbia Ministry of Education’s *History 12 Examination Specifications* (1998) was used. This rubric was not shown to students. Each task was also graded by the classroom teacher. The Pearson correlation between graders (first author and classroom teacher) was .88 for the ill-structured project and .93 for the well-structured project.

Included in this assessment of the ill-structured task was a score out of five for reflective judgment (RJ) based on a grade given independently by the first author. RJ was assessed with close consideration of King and Kitchener’s (1994) seven levels of reflective judgment on ill-structured problems. Pre-reflective learners are in Stages 1, 2, or 3. In Stage 1, learners do not articulate the complexities within an issue (for

which there may be no correct answer), fail to justify their reasoning, and seem to view their opinion as the absolute truth. In Stage 2, learners seem more aware of the dynamics of the issue but they are still dualistic in their justification and over-rely on authorities for truth. Learners in Stage 3 demonstrate some uncertainty and allow for multiple judgments based mainly on personal opinion. At the reflective level (Stages 4 and 5), students demonstrate less certainty but articulate judgements poorly, since they try to accommodate everyone's opinions (Stage 4), before advancing to displaying an awareness that judging an issue requires a contextual and relative stance that accounts for self-limitations. Despite being able to bridge two or more positions they do not coordinate these and the evidence for them well.

To ensure reliability of the RJ score, RJ was graded separately by the classroom teacher and the first author ($r = .89$). To enhance accuracy in assessing RJ, prior to grading the tasks, the classroom teacher and the first author discussed how to assess RJ according to the specifications on the rubric (scored out of 5) and according to King and Kitchener's (1994) RJ model. Since their mean for high school students was 3.2 (out of 7 stages), the rating used in this study assessed students above the pre-reflective level (Stage 4 or higher) with a score of 4–5. Those in Stage 3 were given a 3–4, those in Stage 2 were given a 2–3, and those in Stage 1 were given a 0.5–2 according to the degree to which they oversimplified the issue. Thus, reflective judgment scores were based on levels of awareness demonstrated about external sources of evidence for the complexities and contradictions in the task, on juxtaposing evidence for and against each position (for research or promotion in defeating cancer), and on considering bias in oneself, authors, or sources.

Results

Factor Analyses

To discern the factor structure of epistemological beliefs (EB) within the overall sample, a principal axis exploratory factor analysis with varimax rotation and a minimum eigenvalue of .98 was performed on 52 of 63 MEQ items (justified earlier), extracting 18 factors that accounted for 36.45% of the variance. The internal consistency coefficient for the SMEQ was .67. The first three extracted factors—fixed and quick ability to learn (FQAL), simple knowledge (SK), and certain knowledge (CK)—accounted for 16.81% of the variance (see Table 1). Fixed and quick ability to learn (7.49% of the variance, eigenvalue = 3.89), simple knowledge (6.51%, eigenvalue = 3.38), and certain knowledge (2.82%, eigenvalue = 1.47) had internal consistency reliability coefficients of .65, .51, and .73 respectively. Pearson correlation coefficients (r) between factors were: FQAL and SK .08, FQAL and CK .04, and SK and CK $-.002$. Content validity for these three dimensions of epistemological beliefs is evident through their consistent emergence in previous research with secondary school students (e.g., Qian & Alvermann, 1995; Schommer, 1993; Schommer et al., 1992, 1997; Schommer & Dunnell, 1997). These psychometric properties were deemed suitable to justify the three primary factors being used in subsequent analyses within this study.

Table 1. Combined sample items and factor loadings (alpha)

Factor items	Fixed and quick ability to learn ($\alpha = .65$)	Simple knowledge ($\alpha = .51$)	Certain knowledge ($\alpha = .73$)
The really smart students don't have to work hard to do well in school.	.58		
Successful students understand things quickly.	.55		
Some people are born good learners, others are just stuck with limited ability.	.53		
Working hard on a difficult problem for a longer period of time only pays off for really smart students.	.48		
Once a person is born, their ability to learn will not change.	.39		
Books that teach us how to do something are not much help.		.46	
You will just get confused if you try to take ideas that are new to you from a topic in a textbook and try to mix them with what you already know about that topic.		.43	
We can't be sure of anything, except that we will die someday.		.41	
It's a waste of time to work on problems which have no possibility of coming out with a clear-cut and straight forward answer.		.33	
Scientists can eventually get to the truth.			.76
If scientists try hard enough, they can find the truth to almost anything.			.74
% variance	7.49	6.51	3.38

Note. $n = 447$; eigenvalues $> .98$.

School Orientation Differences

Multivariate analyses of variance ($p < .05$), contrasting the epistemological belief factor scores between samples from the public schools and that from the independent school, showed a significant main effect ($F[3,437] = 4.68, p = .003, \eta^2 = .03$). Post-hoc t tests ($p < .05$) revealed statistical differences between public and independent schools in CK ($t[439] = 3.44, p = .001, \eta^2 = .03$) but not in FQAL ($t[439] = .38, p = .71, \eta^2 < .001$) or SK ($t[439] = 1.57, p = .12, \eta^2 = .006$). Students in the independent (religious) school had more sophisticated beliefs overall, especially in relation to certain knowledge, than students in the public (non-affiliation) schools. There were no differences in EB ($p < .05$) between ethnic groups.

Epistemological Beliefs and Overall Academic Achievement

The means of estimates of overall academic average were 74.11 ($SD = 10.87$) for the combined sample, 75.66 ($SD = 8.90$) for the independent school sub-sample, and

Table 2. Means and standard deviations for factors by group and samples

Group	<i>n</i>	Fixed and quick ability to learn	Simple knowledge	Certain knowledge
Combined sample:				
Total	447	.000 (.84)	.000 (.79)	.000 (.86)
Boys	218	.20 (.85)***	-.06 (.85)	.11 (.88)*
Girls	227	-.19 (.78)***	.06 (.72)	-.10 (.84)*
Low EOAA	123	.21 (.89)	.18 (.85)	-.05 (.89)
Moderate EOAA	198	.005 (.76)	.02 (.77)	.07 (.86)
High EOAA	124	-.22 (.85)	-.21 (.70)	-.06 (.84)
Independent school sub-sample:				
Total	89	-.03 (.78)	-.12 (.84)	-.28 (.82)+
Girls	44	-.09 (.81)	-.006 (.75)	-.37 (.70)
Boys	45	.03 (.75)	-.23 (.91)	-.19 (.93)
Public school sub-sample:				
Total	358	.008 (.85)	.03 (.77)	.07 (.86)+
Girls	184	-.22 (.77)	.08 (.72)	-.04 (.86)
Boys	174	.24 (.87)	-.02 (.83)	.19 (.85)

Note. The higher the score the more naïve the belief. Values enclosed in parentheses represent standard deviations. EOAA = estimated overall academic achievement; + = difference between school orientation ($p < .001$); * = difference between gender within the combined sample ($p < .05$); *** = difference between gender within the combined sample ($p < .001$).

73.73 ($SD = 11.29$) for the public school sub-sample. The strong Pearson correlation ($r = .91$) between EOAA and OAA noted earlier provides evidence for the validity of EOAA as a representation of overall academic achievement. Regression analyses ($p < .05$) were used to determine the strength of each factor in predicting student EOAA in the combined sample. To control for CK (the lone factor to differ according to school orientation), it was entered first in a sequential regression followed by simultaneous entry of FQAL and SK. After partialling out CK ($R^2 = .00$, $F[1,438] = .00$, $p = 1.00$), FQAL ($R^2 = .04$, $F[1,438] = 17.38$, $p < .001$) and SK ($R^2 = .05$, $F[1,438] = 20.48$, $p < .001$) strongly predicted EOAA. The increment in R^2 was .08 ($F[3,436] = 12.09$, $p < .001$). To summarise, the predictors of EOAA are FQAL and SK, with or without controlling for CK. Results of statistically significant regression analyses are presented in Table 3.

To assess differences in EB as a function of low, moderate, and high EOAA in the combined sample, a multivariate analysis of variance ($p < .05$) was performed with levels of EOAA entered as fixed factors while EB factor scores served as dependent variables. There was a significant main effect ($F[6,874] = 5.52$, $p < .001$, $\eta^2 = .04$) and significant differences between levels of EOAA in FQAL ($F[2,438] = 8.52$, $p < .001$, $\eta^2 = .04$) and SK ($F[2,438] = 8.08$, $p < .001$, $\eta^2 = .04$), but not in CK ($F[2,438] = 1.23$, $p = .29$, $\eta^2 = .006$). Post-hoc t tests ($p < .05$) assessing differences in EB as a function of low, moderate, and high EOAA in the combined sample

Table 3. Summary of significant ($p < .05$) regression analyses

Predictor	Outcome	R^2	F	p
Combined sample:				
EB factors	EOAA	.08	12.09***	<.001
FQAL	EOAA	.04	17.38***	<.001
Simple knowledge	EOAA	.05	20.48***	<.001
Independent sub-sample:				
EB factors	IST performance	.12	3.80*	.01
Simple knowledge	IST performance	.06	5.84*	.02
EB factors	Reflective judgment	.10	2.97*	.04
Simple knowledge	Reflective judgment	.06	5.56*	.02

Note. EOAA = Estimates of overall academic achievement; IST = ill-structured task; EB factors = The study's three factors for epistemological beliefs; FQAL = fixed and quick ability to learn.

revealed statistical differences between the high and low EOAA groups in FQAL ($t[242] = 3.89, p < .001, \eta^2 = .06$) and SK ($t[242] = 3.97, p < .001, \eta^2 = .06$), but not in CK ($t[242] = .12, p = .90, \eta^2 < .001$). Differences between the high and moderate EOAA groups were evident in FQAL ($t[318] = 2.46, p = .01, \eta^2 = .02$) and SK ($t[318] = 2.74, p = .007, \eta^2 = .02$), but no difference was noted in CK ($t[318] = 1.39, p = .17, \eta^2 = .006$). Finally, differences between low and moderate EOAA were noted in FQAL ($t[316] = 2.24, p = .03, \eta^2 = .02$) but not in SK ($t[316] = 1.74, p = .08, \eta^2 = .009$) or in CK ($t[316] = -1.22, p = .23, \eta^2 = .005$). Thus, students at different levels of EOAA consistently differ in FQAL and SK but not in CK. Descriptive statistics are presented in Table 2 and Figure 1.

Task Performance

A t test showed that task order did not affect achievement ($p \geq .05$). Disregarding task order, there was no statistical difference between achievement on the well-structured task (WST: $M = 70.30, SD = 17.61$) and on the ill-structured task (IST: $M = 67.72, SD = 17.93; t[88] = -1.38, p = .17, ES = 0.15$). The Pearson correlation coefficient between performance on the IST and performance on the WST was .51.

Epistemological Beliefs, Task Performance, and Reflective Judgment

To determine the strength of each factor in predicting students' performance scores on the ill-structured and well-structured tasks, the three EB factors were entered as a block of predictors in a regression analysis ($p < .05$). The EB factors predicted students' performance on the IST ($R^2 = .12, F[3,85] = 3.80, p = .01$). Individually, SK ($R^2 = .06, F[1,87] = 5.84, p = .02$) was a statistical predictor, but not FQAL ($R^2 = .01, F[1,87] = 1.28, p = .26$) or CK ($R^2 = .03, F[1,87] = 2.64, p = .11$). Since CK was the lone factor to differ according to school orientation, it was controlled by

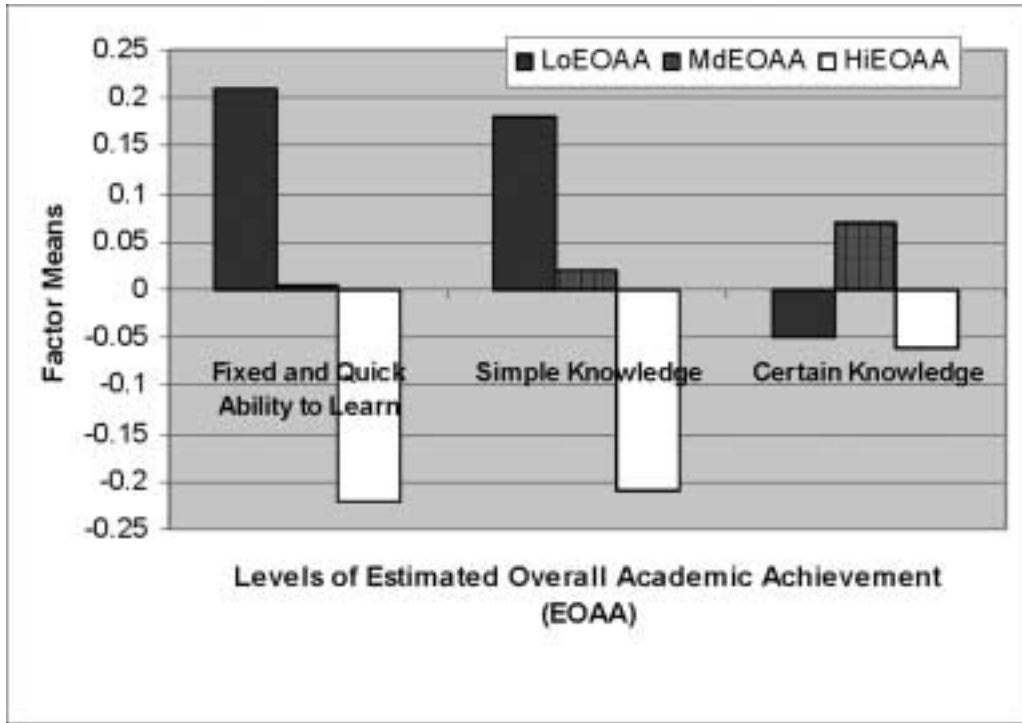


Figure 1. High, moderate, and low estimates of overall academic achievement (EOAA) and mean factor scores ($n = 447$) (note: the higher the score the more naïve the belief)

entering it first in a structured regression followed by SK. After partialling out CK, SK continued to predict performance on the IST (increment in $R^2 = .08$, $F[2,86] = 5.36$, $p = .006$). The EB factors did not predict performance on the well-structured task collectively ($R^2 = .04$, $F[3,85] = 1.07$, $p = .37$) or individually: FQAL ($R^2 = .005$, $F[1,87] = .42$, $p = .52$), SK ($R^2 = .002$, $F[1,87] = .16$, $p = .69$), and CK ($R^2 = .03$, $F[1,87] = 2.55$, $p = .11$).

The same procedure ($p < .05$) discovered significant predictive relations between EB and the RJ score on the ill-structured task ($R^2 = .10$, $F[3,85] = 2.97$, $p = .04$). Individually, SK ($R^2 = .06$, $F[1,87] = 5.56$, $p = .02$) predicted reflective judgment. FQAL ($R^2 = .01$, $F[1,87] = 1.03$, $p = .31$) and CK ($R^2 = .02$, $F[1,87] = 1.36$, $p = .25$) did not. After controlling for CK, SK continued to predict RJ on the IST (increment in $R^2 = .07$, $F[2,86] = 4.20$, $p = .02$). Thus, EB predicted performance on the IST not the WST and, with or without controlling for CK, SK predicted reflective judgments and overall performance on the IST.

Gender Differences

The mean EOAA levels for girls and boys in the combined sample were 76.9 and 72.05. An independent samples t test ($p < .05$) showed that girls had significantly

higher EOAA ($t[444] = 3.99, p < .001$). Girls' mean performances on the tasks were 71.64 (IST) and 75.60 (WST), compared to the boys' 63.88 (IST) and 65.12 (WST). Girls performed significantly better on both tasks (IST: $t[87] = 2.08, p = .04, ES = .44$; WST: $t[87] = 2.92, p = .004, ES = .58$).

To assess differences in EB according to gender in the combined sample, a multivariate analysis of variance ($p < .05$) was performed with gender entered as the fixed factor while EB factor scores served as dependent variables. There was a significant main effect difference in EB across gender ($F[3,437] = 12.82, p < .001, \eta^2 = .08$). Post-hoc t tests ($p < .05$) revealed statistical differences between genders in FQAL ($t[439] = -5.07, p < .001, \eta^2 = .06$) and CK ($t[439] = -2.59, p = .01, \eta^2 = .02$), but not in SK ($t[439] = 1.65, p = .10, \eta^2 = .006$). Thus, girls had statistically higher academic averages, performed better on both tasks, and had significantly more sophisticated beliefs in FQAL and CK than boys.

Discussion

The intentions for this study were to investigate the SMEQ factor structure, differences in epistemological beliefs between public and independent (religious) school samples, and relations between epistemological beliefs and gender, overall academic achievement, reflective judgment, and performances on an ill-structured and a well-structured task. It was specifically posited that epistemological beliefs would have a stronger predictive relationship with performance on an ill-structured task than on a well-structured task, that epistemological beliefs would relate to students' reflective judgment scores, and that girls would achieve significantly higher grades and performance scores on both tasks while believing less in quick learning and fixed ability than boys.

To increase the validity of the SMEQ for use in this study, the recommendations of other researchers (e.g., Qian & Alvermann, 1995; Schommer, 1993) were followed and the main factors resulting from an exploratory factor analysis of items were used. The main factors to emerge were fixed and quick ability to learn (FQAL), simple knowledge (SK), and certain knowledge (CK). The emergence of the FQAL factor lends support to assertions (e.g., Hofer, 2000, 2005) that quick learning and fixed ability may be metacognitive processes generalisable to, or independent from, other more epistemological dimensions. Hofer (2000, 2005) and others (e.g., Hofer & Pintrich, 1997; Pintrich, 2002) explain that beliefs about the speed of learning and about the incremental nature of intelligence are important constructs related to learning, but are not part of a philosophical understanding of a learner's epistemology, which comprises beliefs about knowledge and knowing. Schommer-Aikins (2004) has recently distinguished beliefs about knowing from beliefs about learning. While the debate about whether beliefs about learning and intelligence constitute part of personal epistemology centred on the nature of knowledge and knowing continues to be a worthy one, this study reveals that beliefs about intelligence and the speed of learning (FQAL) and knowledge (SK and CK) are linked to several constructs

affecting achievement in the academic domain and are, therefore, worthy of further analyses regardless of their classification.

In this study, students from the Protestant-Christian school had more sophisticated epistemological beliefs overall, and beliefs about certain knowledge in particular, than students from the public schools. This may contradict other research (e.g., Dale, 1995; Kitchener & King, 1981) that has linked students' beliefs in knowledge as a revelation from God or "absolute authority-based knowledge" (p. 113) with weaker reflective judgment. Since many students in the public school sample are also probably religious, another explanation may be that the curricula of the independent school or the learning environments in the homes of these students may stimulate more critical thinking, thereby fostering more availing epistemological beliefs. Perhaps more students in such schools have acquired a superior capability to coordinate multiple and contrasting views relative to context (Hofer, 2001). More research into the causal nature of such differences according to school orientation would be welcomed.

Students' overall epistemological beliefs and beliefs in FQAL and SK predicted estimates of overall academic achievement, which corroborates other research on epistemological beliefs and achievement (Boyes & Chandler, 1992; Qian & Alvermann, 1995; Schommer, 1993; Schommer et al., 1992; Wood & Kardash, 2002). The finding that beliefs in FQAL predicted academic achievement in this study is noteworthy, since beliefs about innate ability—while having been linked to certain goal orientations, choice of tasks, and cognitive operations—have not been strongly linked to students' academic performance (Qian & Alvermann, 1995) other than in gifted students (Schommer & Dunnell, 1997).

Epistemological beliefs, particularly beliefs related to FQAL and SK, were significantly more naïve in lower than higher academically achieving students. Other research (e.g., Hofer, 1999; Mason & Boscolo, 2002) has linked non-availing epistemological beliefs to various indices of academic achievement—each being part of the tapestry of academic success and helping explain why more availing beliefs in FQAL and SK are held by higher achieving students in this study. Epistemological beliefs may mediate academic learning and performance indirectly, along with other constructs. For example, students who view ability as fixed also have much lower levels of perceived self-efficacy in problem solving, and potentially lower subsequent academic performances (Bandura, 1993).

The determinant strength of epistemological beliefs on task performance was significant for the ill-structured task but not for the well-structured task. This lends some support to other research addressing similar instructional design queries (e.g., Schraw et al., 1995). When tasks are structured loosely to increase their complexity, this may influence task performance by activating epistemic motivations that mediate psychological operations for engaging cognitively in accessing information, developing and testing new theories, persevering through discrepancies, and seeking appropriate closure (Kruglanski, 1990). In instruction, teachers should monitor the cognitive complexity and motivational cost of learning tasks, along with students' beliefs that ability is fixed, that knowledge is simply an accumulation of isolated

facts, that integration is not necessary, and that learning is accomplished the first time. The type of structure of assigned tasks also merits attention if, as Spiro et al. (1988) conjecture, assigning too many well-structured tasks can promote a false understanding that the domain is fundamentally simple and can stimulate superficial engagement. However, balance is important—assigning especially ambiguous tasks may induce anxiety and withdrawal (Doyle, 1983).

The finding that belief in simple knowledge had statistical links to both performance and reflective judgment scores on the ill-structured task corroborates other research relating capabilities for reflective judgment to performance on ill-structured tasks (King & Kitchener, 1994, 2002). Belief in simple knowledge appears to have a particularly influential role on performance on ill-structured tasks, perhaps because of its mediating effect on reflective judgment ability. On the basis of these results, it seems that students who see knowledge as simple—that is, they seek single answers, avoid ambiguity, and depend too much on authority—tend to experience more difficulty with the ambiguous features of tasks that call for reflective judgments, perseverance, and appropriate self-regulated learning. Viewing knowledge as simple can limit subsequent conceptual change and performance in particular academic contexts (Qian & Alvermann, 1995; Schommer & Dunnell, 1997). For instance, Schommer et al. (1992) found that simple knowledge predicted test performance on differently structured maths problems; it was negatively associated with comprehension and metacomprehension, and was mediated by study strategy.

Roth and Roychoudhury (1994) hypothesised that since knowledge structures, epistemological assumptions, and tasks differ across domains, it may be prudent to relate epistemological beliefs to specific contexts rather than to descriptions of student's general views. At certain ages, epistemological beliefs may be too global or unidimensional to differ substantially across achievement levels and tasks, or they may be more elusive to current measurement instruments. Winne and Hadwin (1998) suggest that individual differences in epistemological beliefs that influence perceptions of learning tasks may transcend tasks, thus rendering them unresponsive to task conditions. However, Mason and Boscolo (2004) reported that high school students with more availing epistemological beliefs altered their topic-specific beliefs more than those with moderate or less availing beliefs. Useful research contributions could include developing and implementing more task-specific and domain-specific measures for assessing epistemological beliefs in fine-grained learning contexts, and researching links between epistemological beliefs, academic learning, and performance on very specific types of learning tasks.

Finally, this investigation reported gender differences in epistemological beliefs. As predicted, boys believed more naively than girls that one's ability is predetermined, that learning occurs quickly, and that knowledge is certain. Epistemological beliefs such as these could handicap boys' engagement in learning and performance, and may partially account for their low performance relative to girls on both tasks and their lower overall academic achievement. However, Pintrich (2002) noted that male–female biological status may not sufficiently explain group differences in epistemological thinking because it may mask confounding relations with other

constructs, such as ways of knowing and affect. He recommends that “gender orientation in terms of a masculinity–femininity dimension” might better account for such differences (p. 407). Harter, Waters, and Whitesell (1997) found that more androgynous females did not show maladaptive losses of confidence in school. Buehl et al. (2002) add: “The absence of gender differences when responding to general questions about the nature of knowledge may be because males and females may actually reflect on different types of knowledge (informal versus academic) or different academic domains (e.g., mathematics versus history)” (p. 442).

In conclusion, previous research in the educational and psychological literature has highlighted the asynchronous state of the current conceptualisation of epistemological beliefs while proliferating uncertainties and revealing numerous additional needs for study. The results of this study illuminate important relations surrounding the nature, assessment, and specificity of epistemological beliefs. Refinements in current (e.g., Schommer-Aikins et al., 2002) and new (e.g., Buehl et al., 2002; Schraw, Bendixen, & Dunkle, 2002) measures of epistemological beliefs should stimulate increasingly favourable reliability of the construct and its dimensions that could further clarify relationships between its developmental nature and specificity according to age, domain, and instructional context—particularly in relation to learning tasks, social influences, and the worldviews and beliefs of parents and teachers.

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